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GATE ECE · MICROPROCESSORS · CHAPTER-WISE PYQS

Pin Details of 8085 and Interfacing with 8085

GATE ECE Microprocessors · Previous Year Questions with Answer Key

22	17	1988 – 2026	English
QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

INSIDE THIS PDF

Marks 1	14 questions
Marks 2	8 questions
Answer key	at the end

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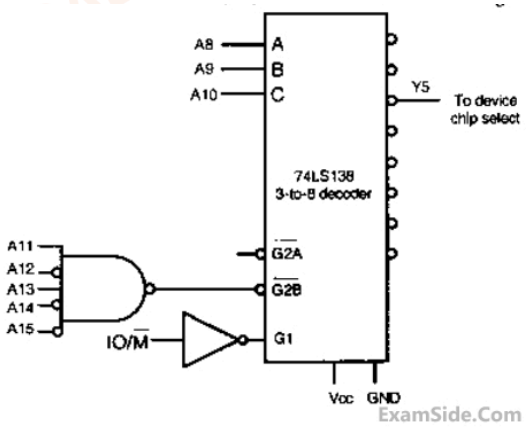
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Marks 1 14 questions

Q1. GATE ECE 2010

In the circuit shown, the device connected to Y_5 can have address in the range

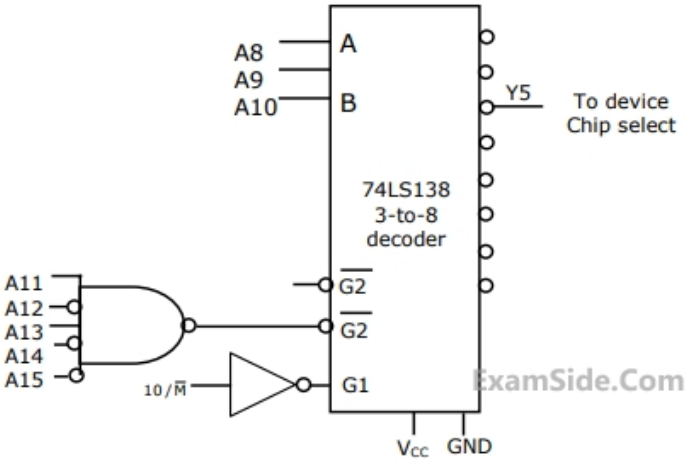


Open solution

(A) 2000H - 20FFH (B) 2D00H - 2DFFH (C) 2E00H - 2EFFH (D) FD00H - FDFFH

Q2. GATE ECE 2010

In the circuit shown, the device connected to Y5 can have address in the range



Open solution

(A) 2000H - 20FFH (B) 2D00H – 2DFFH (C) 2E00H– 2EFFH (D) FD00H- FDFFH

Q3. GATE ECE 2000

The number of hardware interrupts (which require an external signal to interrupt) present in an 8085 microprocessor are

(A) 1 (B) 4 (C) 5 (D) 13

Open solution

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Q4. GATE ECE 2000

In the 8085 microprocessor, the RST6 instruction transfers the program execution to the following location

(A) 30 H (B) 24 H (C) 48 H (D) 60 H

Open solution

Q5. GATE ECE 1999

If $CS = \overline{A_{15}}\ A_{14}\ A_{13}$ is used as the chip select logic of a 4K RAM in an 8085 system, then its memory range will be

(A) 3000H – 3FFFH
(B) 7000H – 7FFFH
(C) 5000H – 5FFFH and 6000H – 6FFFH
(D) 6000H – 6FFFH and 7000H – 7FFFH

Open solution

Q6. GATE ECE 1998

An I/O processor control the flow of information between

(A) cache memory and I/O devices
(B) main memory and I/O devices
(C) two I/O devices
(D) cache and main memories

Open solution

Q7. GATE ECE 1997

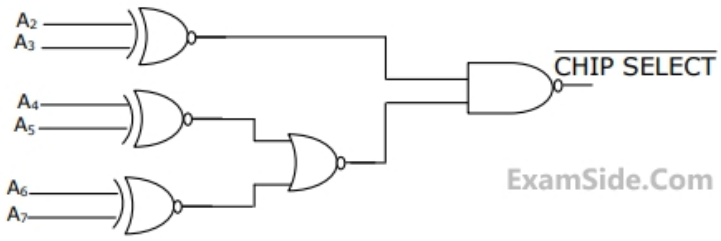
In an 8085μP system, the RST instruction will cause an interrupt

(A) only if an interrupt service routine is not being executed
(B) only if a bit in the interrupt mask is made 0
(C) only if interrupts have been enabled by an EI instruction
(D) None of the above

Open solution

Q8. GATE ECE 1997

The decoding circuit shown below has been used to generate the active low chip select signal for a microprocessor peripheral. (The address lines are designated as A_0 to A_7 for I-O address)



(A) 60 H to H (B) A4 H to A7 H (C) 50 H to AF H (D) 70 H to 73 H

Open solution

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Q9. GATE ECE 1995

In a microprocessor, when a CPU is interrupted, it

(A) Stops execution of instructions.

(B) Acknowledges interrupt and branches of subroutine.

(C) Acknowledges interrupt and continues.

(D) Acknowledges interrupt and waits for the next instrution from the interrupting device.

Open solution

Q10. GATE ECE 1993

In a microcomputer, wait states are used to

(A) make the processor wait during a DMA operation

(B) make the processor wait during an interrupt processing

(C) make the processor wait during a power shutdown

(D) interface slow peripherals to the processor

Open solution

Q11. GATE ECE 1992

In an 8085 microprocessor system with memory mapped I/O,

(A) I/O devices have 16 bit addresses

(B) I/O devices are accessed using IN and OUT instructions

(C) there can be a maximum of 256 input devices and 256 output devices

(D) arithmetic and logic operations can be directly performed with the I/O data.

Open solution

Q12. GATE ECE 1988

For a microprocessor system using I/O-mapped I/O the following statement(s) is NOT true

(A) Memory space available is greater.

(B) Not all data tranfer instructions are available

(C) I/O and Memory address spaces are distinct.

(D) I/O address space is greater.

Open solution

Q13. GATE ECE 1988

A microprocessor with a 16-bit address bus is used in a linear memory selection configuration (i.e. Address bus lines are directly used as chip selects of memory chips) with 4 memory chips. The maximum addressable memory space is

(A) 64k (B) 16k (C) 8k (D) 4k

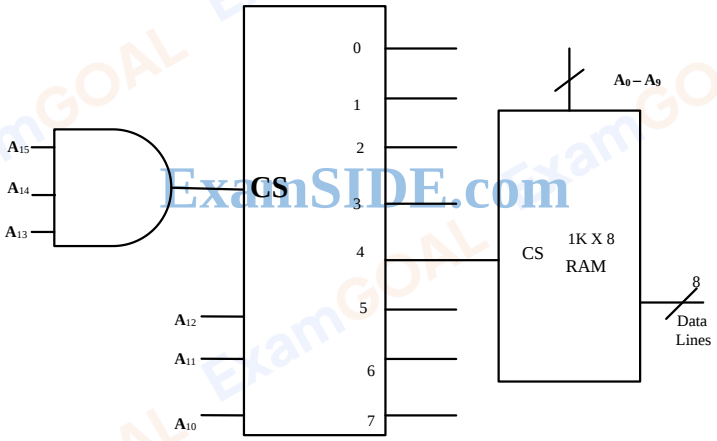
Open solution

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Q14. GATE ECE 1988

A 8bit μp has 16bit address bus. A 1KB memory chip is interfaced to processor as shown ib figure. The address range for the chip is _____.

Open solution



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(A) F000FH - F40EH (B) F100H - F4FFH (C) F000H - F3FFH (D) F700H - FAFFH

Marks 2 8 questions

Q15. GATE ECE 2026

The address of the first location of a 256 kilo byte (KB) memory is $(2500)_{\mathrm{H}}$. Choose the correct address of the last location of the memory.

Open solution

(A) $(2\mathrm{FFF})_{\mathrm{H}}$ (B) $(124\mathrm{FF})_{\mathrm{H}}$ (C) $(424\mathrm{FF})_{\mathrm{H}}$ (D) $(324\mathrm{FF})_{\mathrm{H}}$

Q16. GATE ECE 2016 Set 2

An 8 Kbyte ROM with an active low Chip Select input CS is to be used in an 8085 microprocessor based system. The ROM should occupy the address range 1000H to 2FFFH. Theaddress lines are designated as A15 to A0, where A15is the most significant address bit.Which one of the following logicexpressions will generate the correct $\overline{CS}$ signal for this ROM?

Open solution

(A) $A_{15} + A_{14} + (A_{13}.A_{12} + \overline{A_{13}}.\overline{A_{12}})$

(B) $A_{15}.\ A_{14}.(A_{13} + A_{12})$

(C) $\overline{A_{15}}.\ \overline{A_{14}}.(A_{13}.\overline{A_{12}} + \overline{A_{13}}.A_{12})$

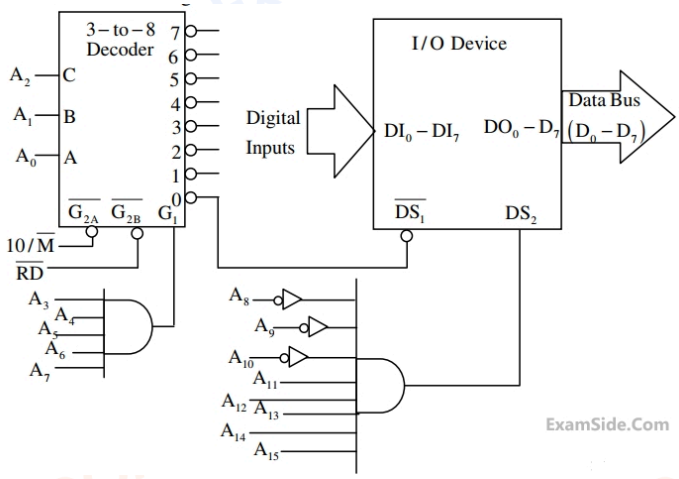
(D) $\overline{A_{15}} + \overline{A_{14}} + A_{13}A_{12}$

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Q17. GATE ECE 2014 Set 2

Open solution

For the 8085 microprocessor, the interfacing circuit to input 8-bit digital data (DI_0 – DI_7) from an external device is shown in the figure. The instruction for correct data transfer is

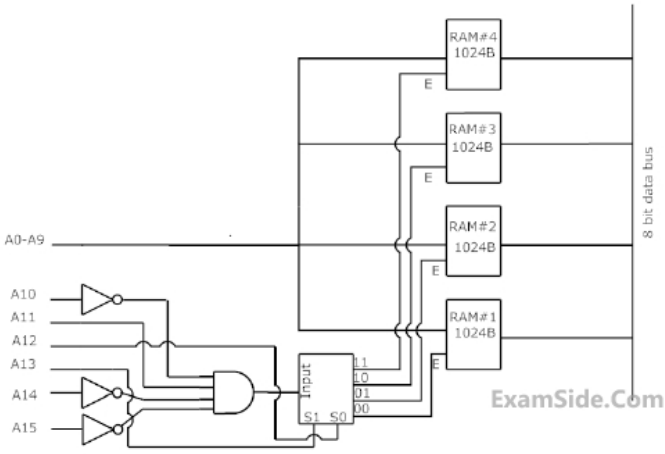


(A) MVI A, F8H (B) IN F8H (C) OUT F8H (D) LDA F8F8H

Q18. GATE ECE 2013

Open solution

There are four chips each of 1024 bytes connected to a 16 bit address bus as shown in the figure below. RAMs 1,2,3 and 4 respectively are mapped to addresses



(A) 0C00H – 0FFFH,1C00H – 1FFFH, 2C00H – 2FFFH,3C00H – 3FFFH

(B) 1800H – 1FFFH,2800H – 2FFFH, 3800H – 3FFFH,4800H – 4FFFH

(C) 0500H – 08FFH,1500H – 18FFH, 3500H – 38FFH,5500H – 58FFH

(D) 0800H – 0BFFH,1800H – 1BFFH, 2800H – 2BFFH,3800H – 3BFFH

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Q19. GATE ECE 2006

An I/O peripheral device shown in figure (b) below is to be interfaced to an 8085 microprocessor. To select the I/O device in the I/O address range D4H – D7H, its chip-select (CS) should be connected to the output of the decoder shown in figure (a) below:

Open solution

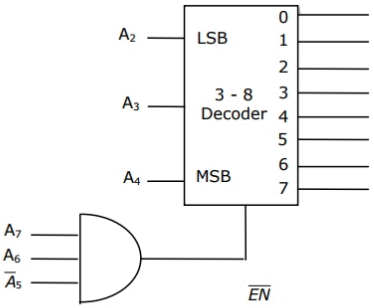
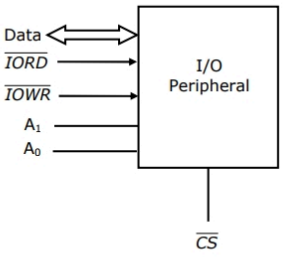


Fig. (a)



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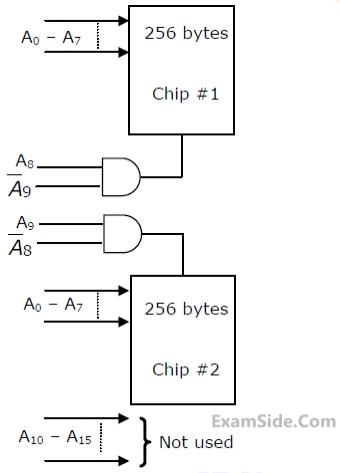
Fig. (b)

(A) Output 7 (B) Output 5 (C) Output 2 (D) Output 0

Q20. GATE ECE 2005

What memory address range is NOT represented by chip 1 and chip 2 in figure? A_0 to A_{15} in this figure are the address lines and CS means Chip Select.

Open solution



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(A) 0100H - 02FFH (B) 1500H - 16FFH (C) F900H - FAFFH (D) F800H – F9FFH

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Q21. GATE ECE 2004

The 8255 Programmable Peripheral Interface is used as described below.
I. An A/D converter is interfaced to a microprocessor through an 8255. the conversion is initiated by a signal from the 8255 on Port C. A signal on Port C causes data to be strobed into Port A.

II. Two computers exchange data using a pair of 8255s. Port A works as a bidirectional data port supported by appropriate handshaking signals.
The appropriate modes of operation of the 8255 for I and II would be

(A) Mode 0 for I and Mode 1 for II
(B) Mode 1 for I and Mode 2 for II
(C) Mode 2 for I and Mode 0 for II
(D) Mode 2 for I and Mode 1 for II

Open solution

Q22. GATE ECE 2001

An 8085 microprocessor based system uses a 4K × 8-bit RAM whose starting address is AA00H. The address of the last byte in this RAM is

(A) AFFFH
(B) B9FFH
(C) BFFFH
(D) A9FFH

Open solution

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Answer Key

Marks 1

1. **D**	2. **B**	3. **C**	4. **A**	5. **D**
6. **B**	7. **C**	8. **A**	9. **D**	10. **D**
11. **A, D**	12. **D**	13. **A**	14. **C**	

Marks 2

15. **C**	16. **A**	17. **D**	18. **D**	19. **B**
20. **D**	21. **B**	22. **B**		

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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